

A SAMPLING DISTRIBUTION

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A Sampling Distribution



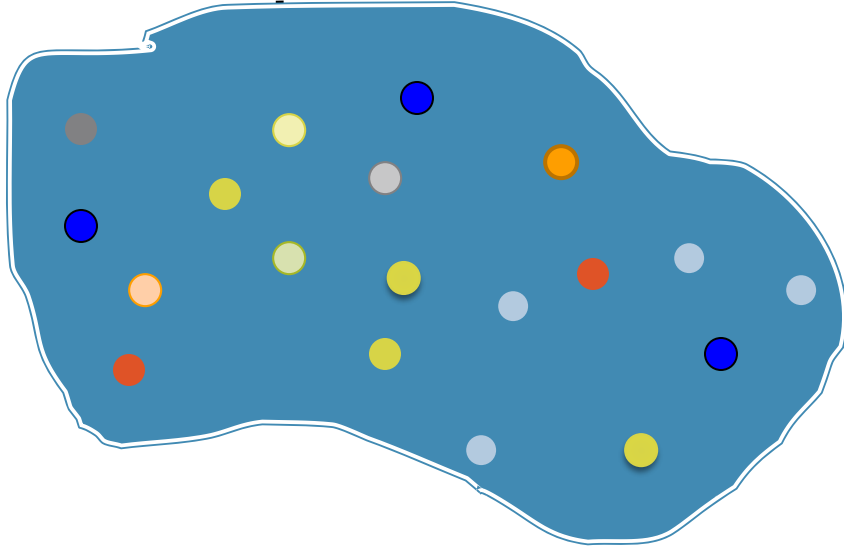
We are moving from descriptive statistics to inferential statistics.



Inferential statistics allow the researcher to come to conclusions about a population on the basis of descriptive statistics about a sample.

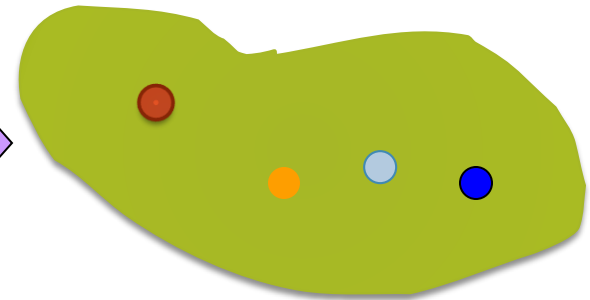
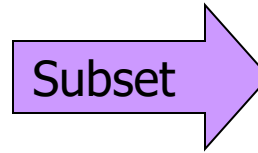
Key Statistical Concepts...

Population



Parameter

Sample



Statistic

- Populations have Parameters,
- Samples have Statistics.

A Sampling Distribution

A theoretical frequency distribution of the scores for or values of a statistic, such as a mean. Any statistic that can be computed for a sample has a sampling distribution.

A sampling distribution is the distribution of statistics that *would be* produced in repeated random sampling (with replacement) from the same population.

We are interested in knowing three things about a given sampling distribution: its *mean*, its *variance*, and its *functional form* (how it looks when graphed).

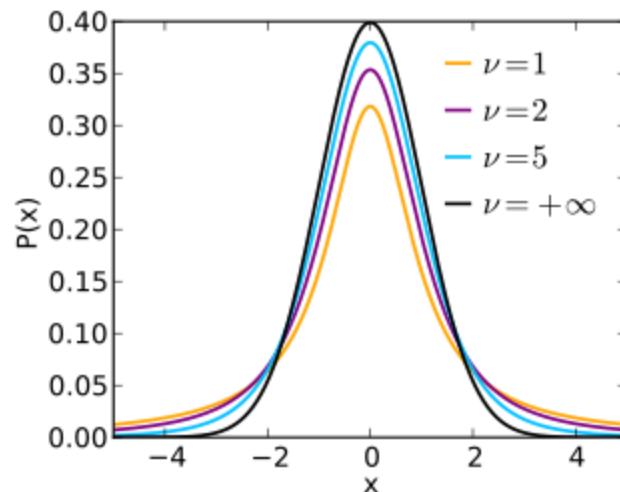
A SAMPLING DISTRIBUTION

The way our means would be distributed if we collected a sample, recorded the mean and threw it back, and collected another, recorded the mean and threw it back, and did this again and again

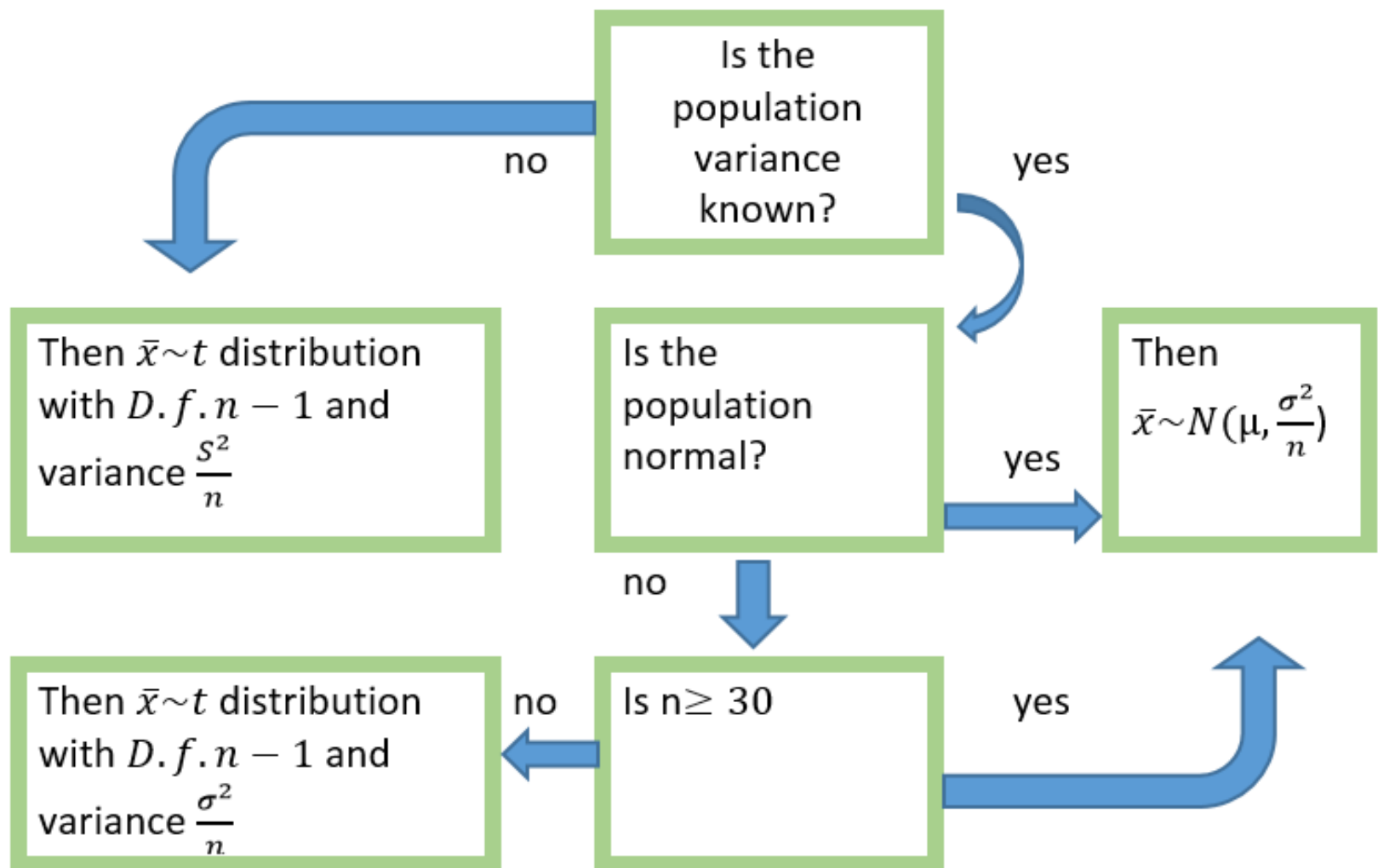
Central limit theorem

- Sampling from Non-normally Distributed Populations, we refer to an important mathematical theorem known as the
- **central limit theorem** : *Given a population of any non-normal distribution the sampling distribution of computed from samples of size n from this population, will have mean μ and variance $(\frac{\sigma^2}{n})$ and will be approximately normally distributed when the sample size is large enough ($n \geq 30$).*

- We need to use sample variance instead of population variance and the sampling distribution is no longer distributed as normal,
- its distributed as t-distribution
- Degrees of freedom $n-1$ with variance $(\frac{s^2}{n})$



A Sampling Distribution



Example

- We took a sample of size 25 from a population that is distributed normal with $\mu=10$ and $\sigma^2=49$, $\sim N(10, 49)$
- The sampling distribution
- $\mu_{\bar{X}}=10$ and $\sigma^2_{\bar{X}}=49/25$
- $\sigma_{\bar{X}}=7/5$
- $\bar{X} \sim N(10, 49/25)$

Example

- We took a sample of size 25 from a population that is distributed normal with $\mu=10$ and σ^2 , $\sim N(10, \sigma^2)$. And the sample variance =49
- The sampling distribution
- $\mu_{\bar{X}}=0$ and $\sigma^2_{\bar{X}}=49/25$
- $\sigma_{\bar{X}}=7/5$
- $\bar{X} \sim t \text{ dist with } d.f., = 25 - 1 = 24 \text{ and } SE = 7/5$

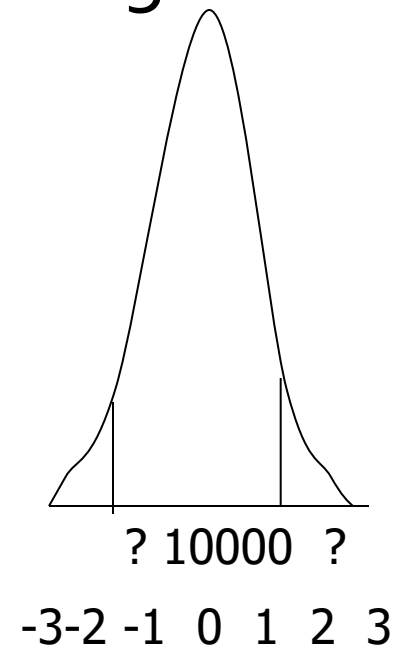
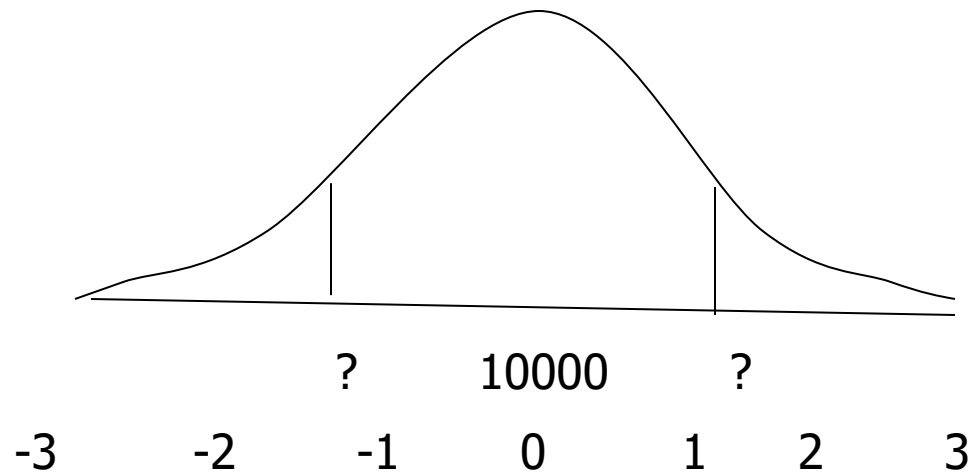
Example

- We took a sample of size 36 from a population that is distributed $\text{Bin}(20, 0.5)$,
- The sampling distribution
- $\mu_{\bar{X}} = np = 10$ and $\sigma^2_{\bar{X}} = np(1-p)/n = 5/36$
- $\sigma_{\bar{X}} = \sqrt{5/36}$
- $\bar{X} \sim N(10, 5/36)$

A Sampling Distribution

An Example:

A population's car values are $\mu = 10,000$ JD with $\sigma = 4,000$. Which sampling distribution figure is for sample size 625 and which is for 2500? What are their s.e.



A Sampling Distribution EXAMPLE

A population's car values are $\mu = 10000$ with $\sigma = 4000$. Which sampling distribution figure is for sample size 625 and which is for 2500? What are their s.e.'s?

$$\text{s.e.} = 4000/25 = 160$$

$$\text{s.e.} = 4000/50 = 80$$

$$(\sqrt{625} = 25)$$

$$(\sqrt{2500} = 50)$$

Which sample will be more precise? If you get a particularly bad sample, which sample size will help you be sure that you are closer to the true mean?

